01 / PROBLEM AND CONTRIBUTION

Breadboard power-supply PCB

Independent hardware design · KiCad | PCB design study

DESIGN AT A GLANCE

Created a KiCad breadboard-supply design with LM7805 and LM317 regulator stages, selectable output headers, and a two-copper-layer board.

Engineering question

How can a compact supply board make regulated power and selectable rails accessible for prototyping?

Situation

The project organizes input power, regulation, switching, indication, and output connectors into a breadboard-oriented module. The schematic labels 5 V and 3.3 V output rails.

Task

Translate the supply circuit into a board with accessible switches, connectors, and regulator placement.

My contribution

Schematic capture, PCB implementation, and design documentation in the linked personal GitHub project.

Tools and concepts

KiCad / PCB layout / Linear regulators / Power distribution / Connector integration

02 / IMPLEMENTATION

Circuit to board

01 / Design step

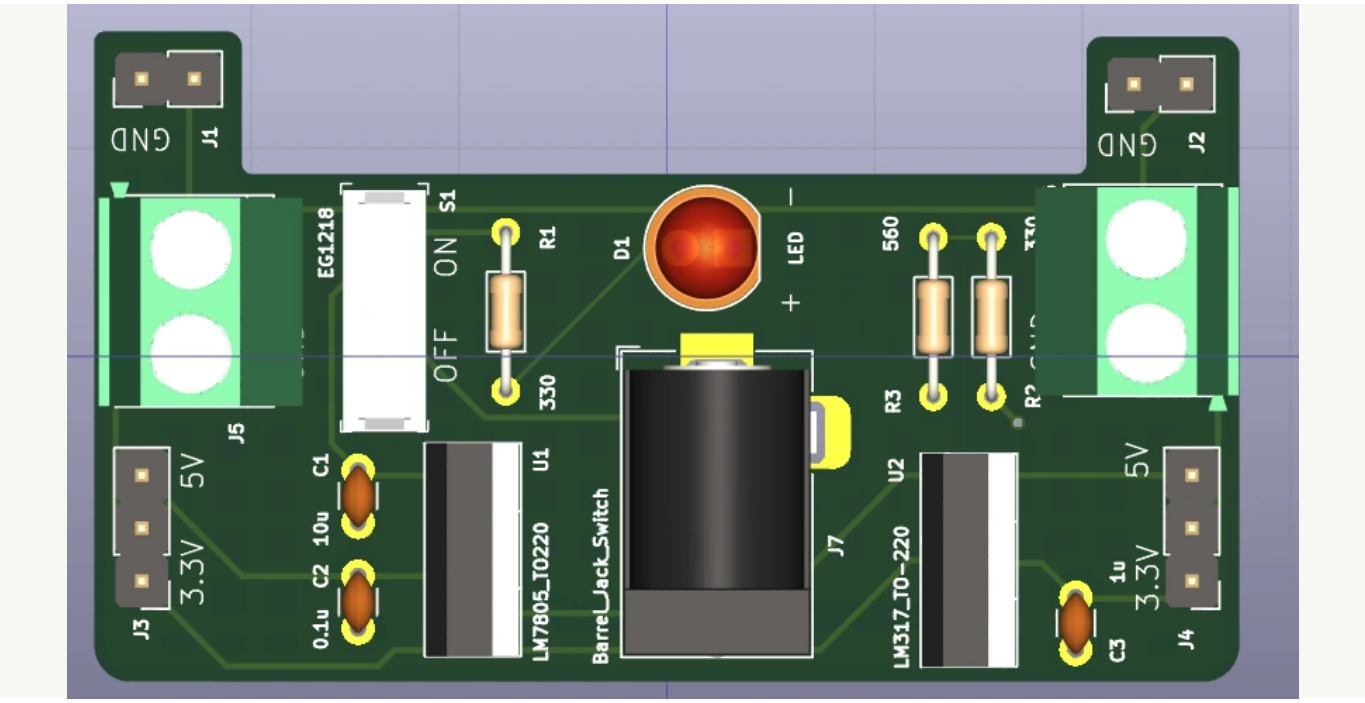
Built separate schematic sections for the input/5 V path, adjustable-regulator path, and output selection. The actual parts are LM7805_TO220 and LM317_TO-220.

02 / Design step

Included input/output capacitors, an LED, switching, and pin/terminal connectors. Retained the schematic's labelled output voltages as design intent, not measured regulation results.

03 / Design step

Implemented the source board with two copper layers, 18 footprints, 63 track segments, and one via, and documented the assembly appearance using a 3D render.



Breadboard-supply board render

Repository 3D render showing regulators, switches, and connectors.

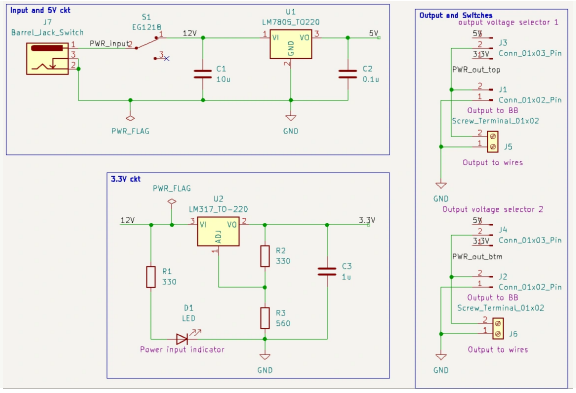
03 / EVIDENCE AND SCOPE

Design evidence and next steps

Current board	2 copper layers
Regulators in source	LM7805 + LM317
Labelled output rails	5 V / 3.3 V

Available evidence

The editable schematic, board file, and matching render show the regulator and connector implementation. The reviewed repository does not provide a load sweep, ripple measurements, or thermal characterization.



Regulation and output-selection schematic

Repository schematic with input, regulator, and selectable-output sections.

Scope and limitations

The README mentions AMS1117 and USB options, whereas the checked-in schematic uses LM7805/LM317 and a barrel-jack input. Output-voltage labels describe intended rails; regulator configuration and loaded behavior require bench verification.

Next steps - proposed work

- Check regulator feedback values and document no-load and loaded output voltages.
- Measure temperature rise and ripple under defined input and load conditions.

Source: PCB-design-projects GitHub repository: Breadboard-Power-Supply; README, KiCad schematic/board, and repository previews. Reviewed 11 November 2026.

Open the project repository